

PART 2: LOGIC SIMPLIFICATION

THEOREMS OF BOOLEAN ALGEBRA

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The animations and contents in these slides are originally adapted from the DLD course slides designed by Dr. Hasan Baig and Mr. Junaid Ahmed Memon. The original slides can be accessed from <https://www.hasanbaig.com/resources>

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POSTULATES OF BOOLEAN ALGEBRA

- If x and y are logic inputs and z is their logical output, then:

- Identity Elements

$$x + 0 = x$$

- Associative Law

$$x + (y + z) = (x + y) + z$$

- Commutative Law

$$x + y = y + x$$

- Complements

$$x + x' = 1$$

- Distributive Law

$$x \cdot (y + z) = (x \cdot y) + (x \cdot z)$$



$$x \cdot 1 = x$$

$$x \cdot (y \cdot z) = (x \cdot y) \cdot z$$

$$x \cdot y = y \cdot x$$

$$x \cdot x' = 0$$

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$



DUALITY PRINCIPLE

- Notice that the Huntington postulates are always listed in pairs. One part may be obtained from the other if the binary operators and the identity elements are interchanged. This important property of Boolean algebra is called the **duality principle**.

Every algebraic expression deducible from the postulates of Boolean algebra remains valid if the operators and identity elements are interchanged.

- If the dual of an algebraic expression is desired, we simply interchange OR and AND operators and replace 1's by 0's and 0's by 1's.



BASIC THEOREMS OF BOOLEAN ALGEBRA

Postulates and Theorems of Boolean Algebra

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$
Theorem 2	(a)	$x + 1 = 1$	(b)	$x \cdot 0 = 0$
Theorem 3, involution		$(x')' = x$		
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Theorem 4, associative	(a)	$x + (y + z) = (x + y) + z$	(b)	$x(yz) = (xy)z$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Theorem 5, DeMorgan	(a)	$(x + y)' = x'y'$	(b)	$(xy)' = x' + y'$
Theorem 6, absorption	(a)	$x + xy = x$	(b)	$x(x + y) = x$



OPERATOR PRECEDENCE

- The operator precedence for evaluating Boolean expressions is:
 1. Parenthesis
 2. NOT
 3. AND
 4. OR



IDEMPOTENCE THEOREM

- Idempotence Theorem (a): $x + x = x$

Statement

$$\begin{aligned}x + x &= (x + x) \cdot 1 \\&= (x + x)(x + x') \\&= x + xx' \\&= x + 0 \\&= x\end{aligned}$$

Justification

postulate 2(b)	$x \cdot 1 = x$
5(a)	$x + x' = 1$
4(b)	$x + yz = (x + y)(x + z)$
5(b)	$x \cdot x' = 0$
2(a)	$x + 0 = x$



IDEMPOTENCE THEOREM

- Idempotence Theorem (b): $x \cdot x = x$

Statement

$$\begin{aligned}x \cdot x &= xx + 0 \\&= xx + xx' \\&= x(x + x') \\&= x \cdot 1 \\&= x\end{aligned}$$

Justification

postulate 2(a)	$x + 0 = x$
5(b)	$x \cdot x' = 0$
4(a)	$x(y + z) = xy + xz$
5(a)	$x + x' = 1$
2(b)	$x \cdot 1 = x$



IDENTITY THEOREM

- Identity Theorem (a): $x + 1 = 1$

Statement

$$\begin{aligned}x + 1 &= 1 \cdot (x + 1) \\&= (x + x')(x + 1) \\&= x + x' \cdot 1 \\&= x + x' \\&= 1\end{aligned}$$

Justification

$$\begin{aligned}\text{postulate 2(b)} \quad & x \cdot 1 = x \\5(a) \quad & x + x' = 1 \\4(b) \quad & x + yz = (x + y)(x + z) \\2(b) \quad & x \cdot 1 = x \\5(a) \quad & x + x' = 1\end{aligned}$$



ACTIVITY 01

Prove the dual of the Identity Theorem

- Identity Theorem (b): $x \cdot 0 = 0$
- Complete the activity to show the above logical statement holds true using previously seen postulates and theorems.

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$



INVOLUTION THEOREM

- Involution Theorem: $(x')' = x$
- From postulate 5, we have $x + x' = 1$ and $x \cdot x' = 0$, which together define the complement of x .
- The complement of x' is x and is also $(x')'$.
- Therefore, since the complement is unique, we have $(x')' = x$.



ABSORPTION THEOREM

- Absorption Theorem (a): $x + x.y = x$

Statement

$$\begin{aligned}x + xy &= x \cdot 1 + xy \\&= x(1 + y) \\&= x(y + 1) \\&= x \cdot 1 \\&= x\end{aligned}$$

Justification

$$\begin{aligned}\text{postulate 2(b)} \quad & x \cdot 1 = x \\4(a) \quad & x(y + z) = xy + xz \\3(a) \quad & x + y = y + x \\2(a) \quad & x + 0 = x \\2(b) \quad & x \cdot 1 = x\end{aligned}$$



ACTIVITY 02

Prove the dual of the Absorption Theorem

- Absorption Theorem (b): $x.(x + y) = x$
- Complete the activity to show the above logical statement holds true using previously seen postulates and theorems.

Postulate 2	(a)	$x + 0 = x$	(b)	$x \cdot 1 = x$
Postulate 3, commutative	(a)	$x + y = y + x$	(b)	$xy = yx$
Postulate 4, distributive	(a)	$x(y + z) = xy + xz$	(b)	$x + yz = (x + y)(x + z)$
Postulate 5	(a)	$x + x' = 1$	(b)	$x \cdot x' = 0$
Theorem 1	(a)	$x + x = x$	(b)	$x \cdot x = x$
Theorem 2	(a)	$x + 1 = 1$	(b)	$x \cdot 0 = 0$



DEMORGAN'S THEOREM

- DeMorgan's Theorem (a): $(x + y)' = x' \cdot y'$
- For any theorem $A = B$, if we can show that $A' \cdot B = 0$, and that $A' + B = 1$, then we will know the statement is true. This proof consists of showing that $(x' \cdot y') \cdot (x + y) = 0$.

$$(x' \cdot y') \cdot (x + y) = 0$$

$$(x' \cdot y') \cdot (x + y) = (x' \cdot y' \cdot x) + (x' \cdot y' \cdot y) \quad \text{by distributive postulate}$$

$$= (x \cdot x' \cdot y') + (x' \cdot y \cdot y') \quad \text{by associativity postulate}$$

$$= 0 \cdot y' + x' \cdot 0 \quad \text{by complement postulate}$$

$$= 0 + 0 \quad \text{by nullity theorem}$$

$$= 0 \quad \text{by identity theorem}$$

$$(x' \cdot y') \cdot (x + y) = 0 \quad \text{proven}$$



DEMORGAN'S THEOREM

- DeMorgan's Theorem (b): $(x \cdot y)' = x' + y'$
- For any theorem $A = B$, if we can show that $A' \cdot B = 0$, and that $A' + B = 1$, then we will know the statement is true. This proof consists of showing that $(x \cdot y) + (x' + y') = 1$.

$$(x \cdot y) + (x' + y') = 1$$

$$\begin{aligned}(x \cdot y) + (x' + y') &= (x + x' + y') \cdot (y + x' + y') && \text{by distributive postulate} \\ &= (x + x' + y') \cdot (y + y' + x') && \text{by associativity postulate} \\ &= (1 + y') \cdot (1 + x') && \text{by complement postulate} \\ &= 1 \cdot 1 && \text{by nullity theorem} \\ &= 1 && \text{by identity theorem}\end{aligned}$$

$$(x \cdot y) + (x' + y') = 1 \quad \text{proven}$$

TRUTH TABLES FOR THEOREM PROOF

Absorption Theorem: $x + x.y = x$

x	y
0	0
0	1
1	0
1	1

xy	$x + xy$
0	0
0	0
0	1
1	1

DeMorgan's Theorem: $(x + y)' = x' . y'$

x	y	$x + y$	$(x + y)'$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

x'	y'	$x' y'$
1	1	1
1	0	0
0	1	0
0	0	0